

SI 305 | Introduction to Information Analysis
 Final Project: Report
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Investigating the Relationship between Housing Age and Elevated Blood Lead Levels in Children in Michigan

1. Summary

Michigan has historically faced challenges related to lead exposure in young children, especially in areas with infrastructure built before the ban on lead in 1978. This project investigated whether Michigan ZIP codes with more pre-1980 housing also showed higher childhood EBLL rates and greater testing coverage from 2010 to 2023. While both relationships were statistically significant, they were weak, suggesting that housing age alone is not a strong predictor of lead risk. As a result, lead screening programs should incorporate additional risk indicators beyond housing age and invest in education campaigns to support testing efforts, especially following Michigan's 2023 universal testing mandate.

2. Problem Statement

Lead exposure remains a long-standing environmental-health concern in Michigan, resulting in many environmental risks for different communities. Studies from the 1980s-1990s identified deteriorating lead-based paint, contaminated dust, and polluted soil as major sources of exposure¹. Michigan's aging housing stock and legacy industrial sites have compounded these risks, continuing to leave children and families vulnerable to lead poisoning across both urban and rural regions². Although federal regulations have reduced new lead emissions, they have not removed existing hazards in older housing and infrastructure. Even after decades of progress, an estimated 500,000 children aged 1 to 5 still exceed the CDC blood lead reference value (BLRV) of 3.5 µg/dL¹.

While anyone is susceptible to lead poisoning, children under six are at greatest risk because their developing bodies absorb lead more easily and their hand-to-mouth behaviors increase ingestion³. Within this age group, exposure risk differs sharply by income, housing age, and race. A national survey found that children most likely to have elevated blood-lead levels live in metropolitan areas, in housing built before 1946, and in low-income African-American and Hispanic families⁴. These differences demonstrate that lead exposure is not evenly distributed but intersects with socioeconomic and racial disparities.

Any level of lead is considered unsafe². Even low concentrations of lead in the bloodstream can result in lower IQ scores and behavioral development⁵. Exposure in early childhood has been linked to

¹ Jonathan Wilson et al., "Pathways and Sources of Lead Exposure: Michigan Children's Lead Determination (the MI CHILD Study)," *Environmental Research* 215 (December 2022): 114204, <https://doi.org/10.1016/j.envres.2022.114204>.

² The Ecology Center, "Michigan's On-Going Lead Poisoning Crisis," *Ecocenter.org*, 2018, <https://www.ecocenter.org/michigans-going-lead-poisoning-crisis>.

³ The Michigan Department of Health and Human Services, "Michigan Annual Blood Lead Levels," March 6, 2025, https://www.michigan.gov/-/media/Project/Websites/mdhhs/Safety-and-Injury-Prevention/Environmental-Health/Mi-Tracking/Documents/Metadata_-_Annual_Blood_Lead_Levels.pdf.

⁴ Michigan League for Public Policy, "A Poison All around Us: The Threat of Lead in Michigan – MLPP," MLPP, March 18, 2016, <http://mlpp.org/a-poison-all-around-us-the-threat-of-lead-in-michigan/>.

⁵ MI Lead Safe, "Lead & Your Health," *Michigan.gov*, 2024, <https://www.michigan.gov/mileadsafe/learn/lead-and-your-health>.

brain damage, hearing and speech problems, and long-term educational challenges⁶. The immediate treatment costs for lead poisoning were estimated at \$271,000 annually, which does not account for the additional expenses associated with managing long-term complications such as learning disabilities, juvenile crime, and special education services⁷. At the community level, persistent lead exposure contributes to wider achievement gaps and deeper intergenerational poverty, magnifying disparities that already exist across Michigan's regions.

The sources of EBLL include deteriorating lead-based paint, corroded pipes, and contaminated soil near industrial areas⁸. Although lead-based paint was banned in 1978, Michigan's aging housing stock remains a major risk factor—nearly 60% of homes in some counties were built before that year⁹. Conditions such as concentrated poverty, poor housing maintenance, and uneven enforcement of safety codes can further exacerbate exposure¹⁰. Until April 2025, testing was not universal. In 2021, only 14% of Michigan children under six were tested, compared with about 40% in Detroit¹¹. There is no single intervention that can fully explain or eliminate exposure; rather, persistent inequities and infrastructure decay sustain the problem across generations.

3. Research Questions

- a. **RQ1:** Are areas in Michigan (ZIP Codes) with a higher share of houses built before 1980 more likely to have children with EBLL from 2010 to 2023?
- b. **RQ2:** Are areas in Michigan (ZIP codes) with a higher share of housing built before 1980 seeing higher rates of childhood EBLL testing from 2010 to 2023?

Note on research questions:

Although both questions examine the share of pre-1980 housing in Michigan ZIP codes, RQ1 focuses on whether housing age is associated with EBLL prevalence, while RQ2 extends the inquiry to testing coverage, asking whether areas with older housing also differ in how extensively children are screened. Together, the questions create a coherent progression—first identifying where risk patterns may appear, and then examining whether testing practices align with those conditions.

⁶ CDC, “Lead Exposure Symptoms and Complications,” Childhood Lead Poisoning Prevention (CDC, April 16, 2024), <https://www.cdc.gov/lead-prevention/symptoms-complications/index.html>.

⁷ Tracy Swinburn, “Costs of Lead Exposure,” October 2016, <https://www.michigan.gov/egle/-/media/Project/Websites/egle/Documents/Programs/DWEHD/Community-Water-Supply/Lead-Copper/cost-of-lead-exposure-and-remediation-in-michigan.pdf?rev=1298f421ff9346a2843b7f5fd9a44f85&hash=234BE184B4C2F651A81F80FDD5B250C8>.

⁸ Jonathan Wilson et al., “Pathways and Sources of Lead Exposure: Michigan Children's Lead Determination (the MI CHILD Study),” *Environmental Research* 215 (December 2022): 114204, <https://doi.org/10.1016/j.envres.2022.114204>.

⁹ Chelsea Devona, “Why Michigan Families Are at Risk of Lead Exposure - Ann Arbor Family,” Ann Arbor Family -, December 8, 2024, <http://annarborfamily.com/health-wellness/why-michigan-families-are-at-risk-of-lead-exposure/>.

¹⁰ Michigan League for Public Policy, “A Poison All around Us: The Threat of Lead in Michigan – MLPP,” MLPP, March 18, 2016, <http://mlpp.org/a-poison-all-around-us-the-threat-of-lead-in-michigan/>.

¹¹ Jennifer Brookland and Clara Hendrickson, “Universal Lead Testing Requirement Landmark Moment for Michigan, Advocates Say,” *Detroit Free Press*, October 3, 2023, <https://www.freep.com/story/news/local/michigan/2023/10/03/new-law-tests-toddlers-for-lead-poisoning/70990292007/?gnt-cfi=1&gca-cat=p&gca-uir=true&gca-epti=z113931d00---v113931b0044xxd004465&gca-ft=160&gca-ds=sophi>.

4. Analysis for RQ1: Is the percentage of pre-1980 housing in Michigan ZIP codes associated with childhood elevated blood lead level (EBLL) rates?

Analysis Method:

I analyzed blood lead testing data from 2010-2023, merged with US Census housing age data across 838 Michigan ZIP codes. The analysis focused on children under age 6 tested at a 3.5 µg/dL blood lead reference value, calculating the average EBLL rate for each ZIP code over the study period. To compare EBLL outcomes across different housing profiles, I grouped ZIP codes into three categories based on the distribution of pre-1980 housing: low (<40%), medium (40–70%), and high (≥70%). I then used Pearson correlation to measure and test the linear relationship between the share of older (pre-1980) housing and EBLL rates. To ensure robust findings, I conducted multiple validation checks, including Spearman rank correlation to test for non-linear relationships, subgroup analysis using the Kruskal-Wallis test to examine whether EBLL rates differed across three housing age categories, and outlier sensitivity analysis using the interquartile range method to verify that extreme values weren't driving the results. Additionally, I tested whether the relationship held when using pre-1970 housing (instead of pre-1980) as an alternative measure and examined whether patterns persisted when analyzing only recent years (2021-2023).

Findings:

There appears to be a weak but statistically significant relationship between the share of pre-1980 housing in a ZIP code and the rate of childhood EBLL in Michigan. Both the correlation analysis ($r = 0.2892$, $p\text{-value} < 0.001$) and the subgroup comparisons across housing-age categories indicate that EBLL rates tend to be higher in areas with older housing, and this pattern holds across multiple robustness checks. The outlier sensitivity analysis also revealed that the findings remain similar after removing outliers (29 ZIP codes or 3.5% of the sample). However, statistical significance and robustness checks do not imply a strong relationship; they only indicate that the observed correlation between housing age and EBLL rates is unlikely to occur by chance. While these results provide useful insight into the research question, several limitations constrain the analysis. A key limitation is that EBLL rates are based only on children who were tested. ZIP codes with older housing may appear to have higher EBLL simply because they test more children, meaning observed differences could reflect detection effort rather than true exposure risk. Another limitation is the use of ZIP-code-level aggregation. Housing conditions, renovation patterns, and socioeconomic factors vary widely within ZIP codes, but the analysis treats each area as uniform. This masks neighborhood-level risk clusters and limits any household-level interpretation. Finally, the housing-age data come from a single Census 5-year estimate, while EBLL data span 2010–2023. Housing stock can change through renovation or redevelopment, and treating it as static may weaken the accuracy of the estimated relationship, particularly in ZIP codes experiencing substantial change over the study period.

Visualizations:

Figure 1: Relationship Between Housing Age and Childhood Lead Exposure Rates by ZIP Code (n = 838)

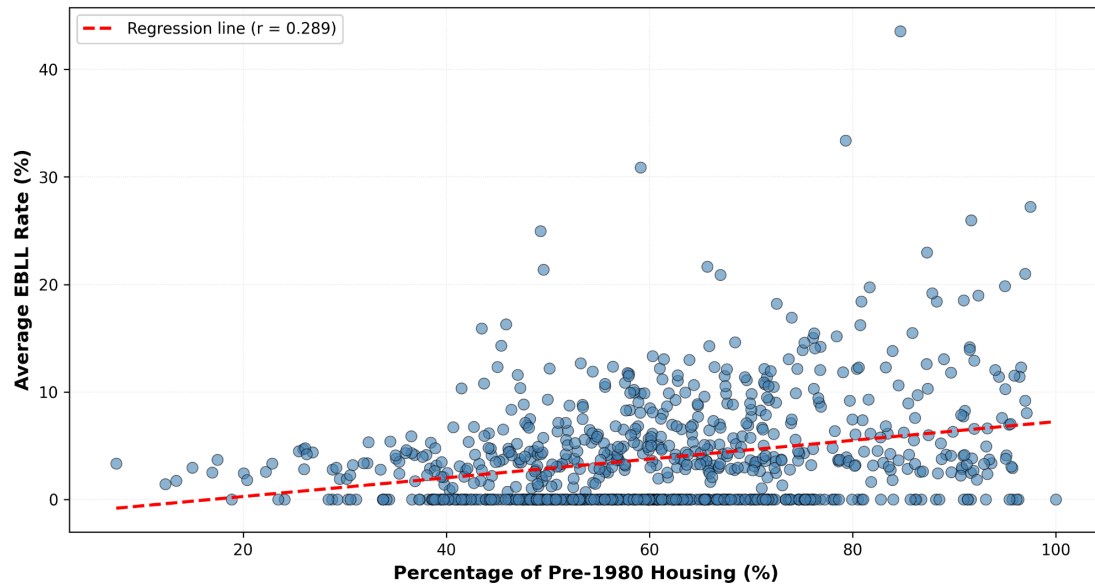


Figure 1. Scatter plot showing the relationship between the share of pre-1980 housing (%) in Michigan ZIP codes (independent variable) and the average EBLL rate (%) among children under 6 (dependent variable). Each point represents a ZIP code.

Figure 2. Elevated Blood Lead Level (EBLL) Rates by Housing Age Category Michigan ZIP Codes (n = 838)

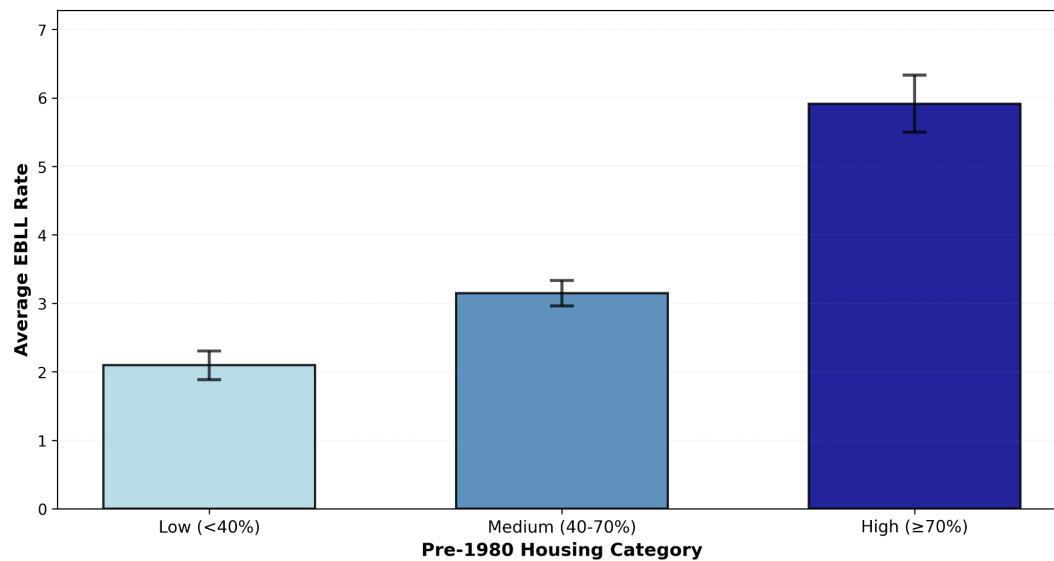


Figure 2. Bar chart displaying the average EBLL rate (%) among children under 6 across three categories of pre-1980 housing: low (<40%), medium (40-70%), and high (≥70%). Error bars represent standard error.

Interpretation

The analysis suggests that Michigan ZIP codes with a higher share of pre-1980 housing tend to have higher average childhood EBLR rates. The observed correlation is statistically significant but weak (Figure 1), indicating that while the association exists, housing age alone does not explain most of the variation in lead exposure. This pattern is reinforced by subgroup comparisons, where ZIP codes with high-housing-age areas had nearly triple the EBLR rate of low-housing-age areas, as seen in Figure 2. This outcome is somewhat aligned with expectations, as housing age is a known risk factor, but it also suggests that other environmental, social, or infrastructural variables likely play major roles. From a policy standpoint, the findings don't suggest an immediate crisis in all older-housing ZIP codes, but they do signal a structural risk pattern that public health agencies should not ignore. Stakeholders such as the Michigan Department of Health and Human Services, municipal housing inspectors, and school districts could use these results to inform more targeted outreach, funding allocation, or enforcement of lead-safe housing regulations. Even a weak but persistent association may justify focused interventions in ZIP codes with homes built before 1980 and higher observed EBLR rates. These findings support the relevance of RQ2 by highlighting the importance of examining whether screening resources are being targeted effectively, especially in structurally vulnerable areas.

5. Analysis for RQ2

Analysis Method:

To examine whether blood lead testing resources align with housing age patterns, I calculated testing coverage rates for 523 Michigan ZIP codes by dividing the average annual number of children tested for lead (2010–2023) by the ZIP-code-level population of children under age 6, using the 2019–2023 American Community Survey 5-Year Estimates. This metric (testing coverage) represents the proportion of young children screened for lead and accounts for differences in child population size across ZIP codes, offering a more meaningful comparison than raw testing counts. I merged this testing coverage data with the pre-1980 housing data and used Pearson correlation analysis to evaluate the linear association between housing age and testing intensity. To validate these findings, I conducted Spearman correlation analysis to check for non-linear patterns, Kruskal–Wallis tests to assess whether average testing coverage differed across ZIP codes in low (<40% pre-1980 housing), medium (40–70% pre-1980 housing), and high ($\geq 70\%$ pre-1980 housing) housing age categories, and outlier sensitivity analysis using the interquartile range method to ensure results were not driven by extreme cases. Additional robustness checks included restricting the analysis to ZIP codes with at least 100 children under age 6, to reduce instability from small population estimates. These combined methods mirror the validation strategy used in the RQ1 analysis and allow for a consistent and rigorous approach to answering RQ2.

Findings:

There appears to be a weak but statistically significant relationship between the share of pre-1980 housing in a ZIP code and the percentage of young children tested for blood lead in Michigan. Both the correlation analysis ($r = 0.347$, $p < 0.001$) and subgroup comparisons across housing-age categories indicate that ZIP codes with older housing tend to have higher testing coverage rates. This pattern was supported across robustness checks, including Spearman correlation, population-size filtering. Outlier sensitivity analysis also confirmed the stability of the findings: after removing 33 ZIP codes (6.3%) with extremely high coverage rates, the correlation increased to $r = 0.425$, suggesting the relationship is slightly stronger when statistical outliers are excluded. However, as with the RQ1 analysis, statistical

significance and robustness do not imply a strong association; the observed relationship is modest in strength and reflects broad trends rather than precise predictive power. The results are limited by the datasets, where testing coverage rate calculation combines blood lead testing data from 2010–2023 with child population estimates from the 2019–2023 American Community Survey, creating a potential mismatch in ZIP codes where the under-6 population has changed significantly over time. This may cause the calculated coverage rates to be slightly inflated or deflated, depending on local population trends. Second, the analysis includes only 523 ZIP codes because over 300 ZIP codes lacked population data for children under six. These missing ZIP codes may represent smaller or more rural areas, which means the findings may better represent trends in mid-size and urban areas than in Michigan’s smallest communities. Together, these factors mean the testing coverage rate is best viewed as an estimate of testing effort—not a precise count of children screened. This limits how precisely RQ2 can assess whether testing resources are aligned with structural risk.

Visualizations:

**Figure 3. Relationship Between Housing Age and Blood Lead Testing Coverage
(n = 523)**

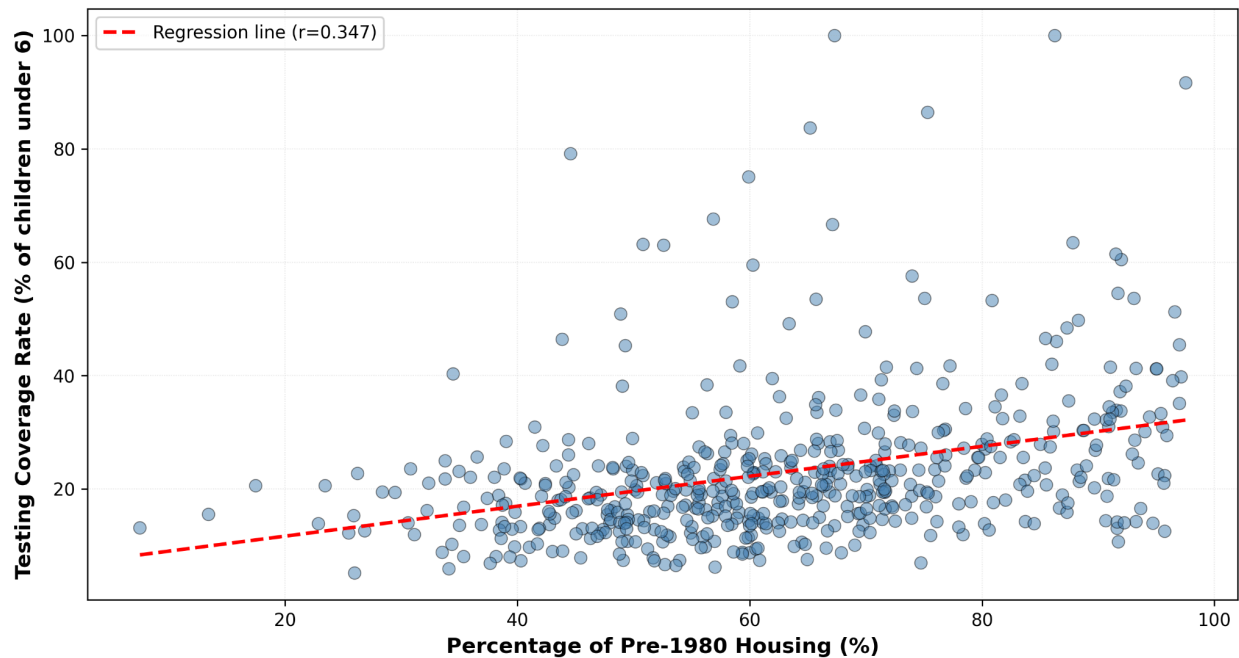


Figure 3. Scatter plot showing the relationship between the share of pre-1980 housing (%) in Michigan ZIP codes (independent variable), and the percentage of children under age 6 tested for EBL (dependent variable). Each point represents a ZIP code.

**Figure 4. Blood Lead Testing Coverage Rate by Housing Age Category
Michigan ZIP Codes (n = 523)**

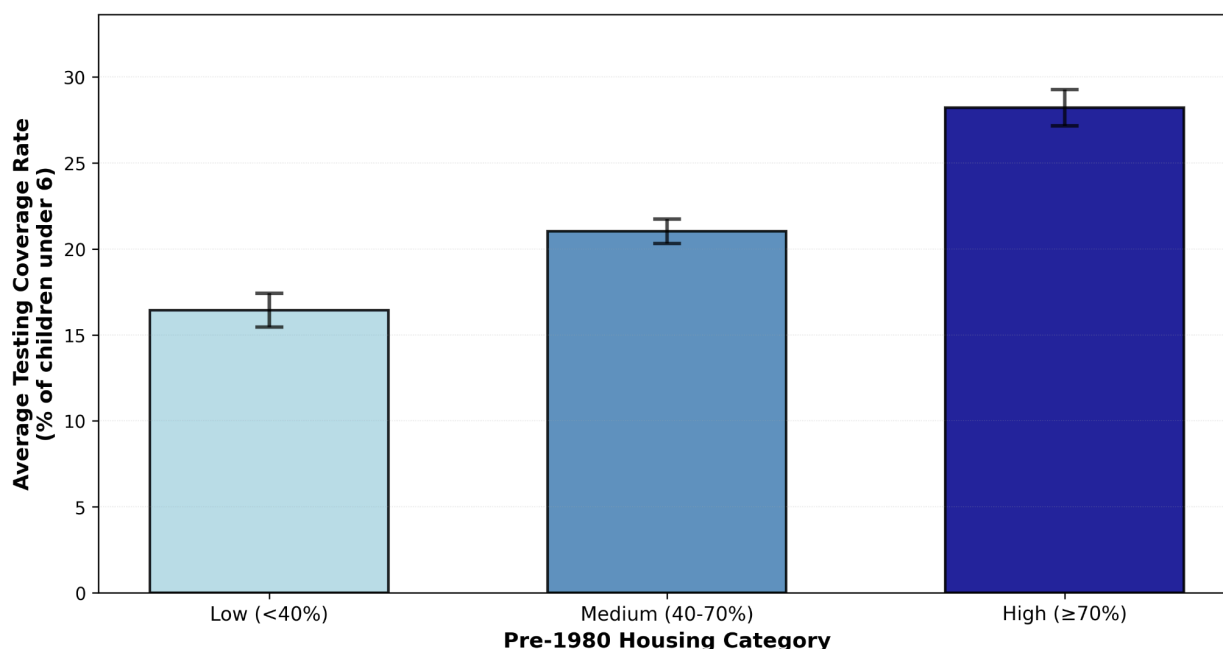


Figure 4. Bar chart displaying the average percentage of children under age 6 tested for EBLL across the three categories of pre-1980 housing: low (<40%), medium (40–70%), and high (≥70%). Error bars represent standard error.

Interpretation:

It appears that Michigan ZIP codes with a higher share of pre-1980 housing tend to test a greater percentage of their young children for elevated blood lead levels. Although the correlation was weak, it still suggests that communities with older housing received somewhat more screening attention prior to 2023, aligning with expectations given the long-recognized link between aging housing stock and lead risk. Subgroup analysis reinforces this pattern, showing a meaningful difference in testing coverage across housing categories. While the relationship is modest, it indicates that housing age likely influenced how testing resources were distributed before the statewide universal testing mandate took effect. Policymakers can use these findings to assess where testing was historically concentrated and identify ZIP codes that may have been under-screened despite potential risk factors. The uneven coverage also suggests that universal testing alone may not guarantee equitable participation; some communities may still need additional outreach, education, or support to ensure families follow through with screening. Local health departments could act on this by targeting ZIP codes with low past testing rates for communication efforts, school-based events, or pediatric provider partnerships. Finally, because this analysis shows correlation rather than causation, these patterns should be interpreted as indicators of where testing occurred—not evidence that older housing directly drives lead exposure or testing behavior. Still, the trends offer a valuable baseline for evaluating how the universal testing policy changes coverage going forward.

6. Conclusion and Future Recommendations

Recommendation 1: Broaden Risk Assessment Criteria Beyond Housing Age

Audience: Michigan Department of Health and Human Services (MDHHS), Local Public Health Agencies

This analysis found that pre-1980 housing is only a weak predictor of EBLL rates and testing coverage, meaning housing age alone cannot reliably identify where children are most at risk. Therefore, MDHHS and local health agencies should revise their lead risk models to incorporate additional indicators, such as ZIP-code-level poverty rates, rental housing turnover, historical EBLL case data, and infrastructure conditions like water service line material. These variables are already available through state databases and could be integrated into updated screening risk maps or planning tools. Doing so would help improve the targeting of lead screening outreach, remediation funding, and follow-up services—particularly in areas where lead exposure risk is high but housing age alone would not flag the need. This recommendation builds on this research question 1’s findings by acknowledging the limitations of a single-variable approach and offering a more comprehensive, data-informed method of allocating public health resources.

Recommendation 2: Expand Community Education Campaigns to Support Universal Lead Testing Compliance

Audience: Michigan Department of Health and Human Services (MDHHS), Local Health Departments

While research question 2 revealed that ZIP codes with older housing showed somewhat higher testing coverage before 2023, wide testing gaps remained across the state, especially in areas with newer housing that may still carry other sources of exposure risks. With Michigan’s universal lead testing mandate going into effect in 2023, increasing awareness and follow-through will determine whether the mandate actually reaches all children. MDHHS and local health departments should run targeted education campaigns in ZIP codes with historically low testing coverage, especially in neighborhoods with linguistic, economic, or logistical barriers. Messaging should be delivered through pediatric clinics, schools, WIC offices, and community centers, and should clearly explain why testing matters and how families can access it. This recommendation stems from RQ2’s finding that testing patterns varied substantially across ZIP codes and that policy changes must be paired with accessible communication to achieve equitable participation.

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